

Ishwor Chalise

Workshop 7

```

ishwor@OHH:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [OHH]
ishwor@OHH:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
ishwor@OHH:~$ jps
1792 NodeManager
2240 Jps
1202 DataNode
1076 NameNode
1670 ResourceManager
1435 SecondaryNameNode
ishwor@OHH:~$

```

Prerequisites and Starting Hadoop Services

```

ishwor@OHH:~$ cd bigdata
ishwor@OHH:~/bigdata$ ls
ishwor@OHH:~/bigdata$ nano $HADOOP_HOME/etc/hadoop/mapred-site.xml
ishwor@OHH:~/bigdata$ nano $HADOOP_HOME/etc/hadoop/mapred-site.xml

```

```

GNU nano 6.2 /usr/local/hadoop/etc/hadoop/mapred-site.xml
<configuration>
  <property>
    <name>yarn.app.mapreduce.am.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
  <property>
    <name>mapreduce.map.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
  <property>
    <name>mapreduce.reduce.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
  <property>
    <name>mapreduce.application.classpath</name>
    <value>/usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/*:/usr/local/hadoop
  </property>
</configuration>

```

Hadoop Directory Structure and Commands

Spark

```
>>> df.show()
+---+-----+-----+---+
|age|course|   lives|name|
+---+-----+-----+---+
| 22|    CS|Kathmandu|Alice|
| 20|    IT|  Pokhara|  Bob|
| 23|    CS|Kathmandu|Carol|
| 21|    IT| Lalitpur|David|
| 24|    CS|  Pokhara|  Eve|
+---+-----+-----+---+

>>> |
```

```
ma()
df.select("name").show()
df.select(df['name'], df['age'] + 1).show()
df.filter(df['age'] > 21).show()
df.groupBy("course").count().show()>>> df.show()
+---+-----+-----+---+
|age|course|   lives|name|
+---+-----+-----+---+
| 22|    CS|Kathmandu|Alice|
| 20|    IT|  Pokhara|  Bob|
| 23|    CS|Kathmandu|Carol|
| 21|    IT| Lalitpur|David|
| 24|    CS|  Pokhara|  Eve|
+---+-----+-----+---+

>>> df.select(df['name'], df['age'] + 1).show()
+---+-----+
| name|(age + 1)|
+---+-----+
|Alice|      23|
|  Bob|      21|
|Carol|      24|
|David|      22|
|  Eve|      25|
+---+-----+

>>> df.filter(df['age'] > 21).show()
+---+-----+-----+---+
|age|course|   lives|name|
+---+-----+-----+---+
| 22|    CS|Kathmandu|Alice|
| 23|    CS|Kathmandu|Carol|
| 24|    CS|  Pokhara|  Eve|
+---+-----+-----+---+

>>> df.groupBy("course").count().show()
```

```

ishwor@OHH:~/bigdata$ nano $HADOOP_HOME/etc/hadoop/mapred-site.xml
ishwor@OHH:~/bigdata$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
ishwor@OHH:~/bigdata$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [OHH]
ishwor@OHH:~/bigdata$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [OHH]
ishwor@OHH:~/bigdata$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
ishwor@OHH:~/bigdata$ jps
4257 NodeManager
3894 SecondaryNameNode
4650 Jps
3659 DataNode
3532 NameNode
4124 ResourceManager

```

```

>>> df.groupBy("course").count().show()
+-----+-----+
|course|count|
+-----+-----+
|    IT|    2|
|    CS|    3|
+-----+-----+

```

```

>>> df.createOrReplaceTempView("student")
name, age, course FROM student WHERE age > 21")
sqlDF.show()>>> sqlDF = spark.sql("SELECT name, age, course FROM student WHERE
> 21")
>>> sqlDF.show()
+-----+-----+
| name|age|course|
+-----+-----+
|Alice| 22|    CS|
|Carol| 23|    CS|
|  Eve| 24|    CS|
+-----+-----+
>>> |

```

```

>>> # SQL version
>>> spark.sql("SELECT name, lives FROM student").show()
+-----+-----+
| name|    lives|
+-----+-----+
| Alice|Kathmandu|
|  Bob|  Pokhara|
| Carol|Kathmandu|
| David|  Lalitpur|
|  Eve|  Pokhara|
+-----+-----+

>>>
>>> # 2. Count how many people live at each place
>>> df.groupBy("lives").count().show()
+-----+-----+
|    lives|count|
+-----+-----+
|  Pokhara|    2|
| Lalitpur|    1|
|Kathmandu|    2|
+-----+-----+

>>> # SQL version
>>> spark.sql("SELECT lives, COUNT(*) as count FROM student GROUP BY lives").show()
+-----+-----+
|    lives|count|
+-----+-----+
|  Pokhara|    2|
| Lalitpur|    1|
|Kathmandu|    2|
+-----+-----+

>>> |

```

```

>>>
>>> # Filter with pipelines
>>> df_weather.filter(df_weather['lang'] == "en").select('user.screen_name', 'user.
location').show(40)
+-----+-----+
|screen_name|location|
+-----+-----+
|   john_doe|    NYC|
| jane_smith|  London|
|  sun_lover|   Miami|
+-----+-----+

>>>
>>> # Pattern matching
>>> df_weather.select('text').filter(df_weather['text'].contains("sun")).show(10, F
alse)
+-----+
|text|
+-----+
|sunny day today|
|cloudy with a chance of sun|
|lots of sun here|
+-----+

>>> df_weather.createOrReplaceTempView("weather")
ark.sql("SELECT user.screen_name, user.location FROM weather WHERE lang = 'en'").sh
ow(50)
spark.sql("SELECT lang, count(*) AS language_count FROM weather GROUP BY lang").sho
w()>>> spark.sql("SELECT user.screen_name, user.location FROM weather WHERE lang =
'en'").show(50)
+-----+-----+
|screen_name|location|
+-----+-----+
|   john_doe|    NYC|
| jane_smith|  London|
|  sun_lover|   Miami|
+-----+-----+

```

```
>>> dfPop.show()
+-----+-----+
| county|year|population|
+-----+-----+
|Wolverhampton|2020| 250000|
|Wolverhampton|2021| 255000|
|Walsall|2020| 280000|
|Walsall|2021| 285000|
|Birmingham|2020| 1100000|
|Birmingham|2021| 1120000|
+-----+-----+

>>> dfPop.select("county").show()
+-----+
| county|
+-----+
|Wolverhampton|
|Wolverhampton|
|Walsall|
|Walsall|
|Birmingham|
|Birmingham|
+-----+

>>> dfPop.filter(dfPop['county'] == 'Wolverhampton').show()
+-----+-----+
| county|year|population|
+-----+-----+
|Wolverhampton|2020| 250000|
|Wolverhampton|2021| 255000|
+-----+-----+

>>> |
```

```
>>>
>>> # Aggregation
>>> sqlDF = spark.sql("""
... SELECT county, count(*) as pop_count, SUM(population) as pop_sum
... FROM pop GROUP BY county ORDER BY county
... """)
>>> sqlDF.show(20, False)
+-----+-----+-----+
|county|pop_count|pop_sum|
+-----+-----+-----+
|Birmingham|2|2220000.0|
|Walsall|2|565000.0|
|Wolverhampton|2|505000.0|
+-----+-----+-----+
```

```
Using Python version 3.12.3 (main, Nov 6 2025 13:44:16)
Spark context Web UI available at http://10.255.255.254:4040
Spark context available as 'sc' (master = local[*], app id = local-1776320665113).
SparkSession available as 'spark'.
>>> text_file = sc.textFile("file:///home/ishwor/input_word.txt")
>>> counts = text_file.flatMap(lambda line: line.split(" ")) \
...     .map(lambda word: (word, 1)) \
...     .reduceByKey(lambda a, b: a + b)
>>> counts.collect()>>> counts = text_file.flatMap(lambda line: line.split(" ")) \
...     .map(lambda word: (word, 1)) \
...     .reduceByKey(lambda a, b: a + b)
>>> counts.collect()
[('hello', 2), ('world', 1), ('spark', 1), ('pyspark', 1), ('workshop', 1)]
>>> counts.saveAsTextFile("file:///home/ishwor/spark_output_word")
```

```
ishwor@OHH:~$ cat ~/spark_output_word/part-*
('hello', 2)
('world', 1)
('spark', 1)
('pyspark', 1)
('workshop', 1)
ishwor@OHH:~$ |
```

Compiling and Running a Custom WordCount Program

Part 4: WordCount MapReduce v1

9. Set up project folder + copy Java file:

```
GNU nano 7.2 WordCount.java *
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {
    public static class TokenizerMapper extends Mapper<Object, Text, IntWritable>{
        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
            StringTokenizer itr = new StringTokenizer(value.toString());
            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class IntSumReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
        private IntWritable result = new IntWritable();
        public void reduce(Text key, Iterable<IntWritable> values, Context context) throws IOException, InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
            result.set(sum);
            context.write(key, result);
        }
    }

    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "word count");
        job.setJarByClass(WordCount.class);
        job.setMapperClass(TokenizerMapper.class);
        job.setCombinerClass(IntSumReducer.class);
        job.setReducerClass(IntSumReducer.class);
    }
}

ishwor@OHH:~/wordcount-v1$ nano WordCount.java
ishwor@OHH:~/wordcount-v1$ javac -classpath $(hadoop classpath) WordCount.java
ishwor@OHH:~/wordcount-v1$ jar cf wordcount.jar Word*.class
ishwor@OHH:~/wordcount-v1$ hdfs dfs -mkdir -p input_word
ishwor@OHH:~/wordcount-v1$ echo "A long time ago in a galaxy far far away" > testfile1
ishwor@OHH:~/wordcount-v1$ echo "Another episode of Star Wars" > testfile2
ishwor@OHH:~/wordcount-v1$ hdfs dfs -put testfile1 testfile2 input_word
ishwor@OHH:~/wordcount-v1$ hdfs dfs -ls input_word
Found 2 items
-rw-r--r-- 1 ishwor supergroup 41 2026-04-16 04:56 input_word/testfile1
-rw-r--r-- 1 ishwor supergroup 29 2026-04-16 04:56 input_word/testfile2
ishwor@OHH:~/wordcount-v1$
```

```
ishwor@OHH:~/wordcount-v1$ hdfs dfs -cat output_word/part-r-00000
A          1
Another    1
Star       1
Wars       1
a          1
ago        1
away       1
episode    1
far        2
galaxy     1
in         1
long       1
of         1
time       1
```

```
ishwor@OHH:~/wordcount-v1$ stop-yarn.sh
Stopping nodemaaagers
Stopping resourcemanager

ishwor@OHH:~/wordcount-v1$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [LAPTOP-EKBESLNE]

ishwor@OHH:~/wordcount-v1$ jps
6912 Jps

ishwor@OHH:~/wordcount-v1$
```

Part 5 – WordCount Version 2 (Improved WordCount)

```
2026-06-16 05:05:58.296 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-06-16 05:05:58.339 INFO impl.MetricsSystemImpl: Scheduled metrics snapshot period at 10 second(s).
2026-06-16 05:05:58.390 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-06-16 05:05:58.406 WARN mapreduce.JobSubmitter: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-06-16 05:05:58.938 INFO input.FileInputFormat: Total input files to process : 2
2026-06-16 05:05:58.982 INFO mapreduce.JobSubmitter: number of splits:2
2026-06-16 05:05:59.252 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local602077923_0001
2026-06-16 05:05:59.270 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-06-16 05:05:59.537 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-06-16 05:05:59.542 INFO mapreduce.Job: Running job: job_local602077923_0001
2026-06-16 05:05:59.546 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-06-16 05:05:59.560 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-16 05:05:59.572 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-16 05:05:59.574 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup, temporary folders under output directory:false, ignore cleanup failures: false
2026-06-16 05:05:59.579 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-06-16 05:05:59.680 INFO mapred.LocalJobRunner: Starting task: attempt_local602077923_0001_n_000000_0
2026-06-16 05:05:59.743 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-16 05:05:59.742 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-16 05:05:59.743 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup, temporary folders under output directory:false, ignore cleanup failures: false
2026-06-16 05:05:59.790 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-06-16 05:05:59.790 INFO mapred.MapTask: Processing split: hdfs://localhost:8020/user/sakshin/input_word/textfile1:0+01
2026-06-16 05:05:59.823 INFO mapred.MapTask: (SQUAD00) @ kvi 26214396(104837584)
2026-06-16 05:05:59.823 INFO mapred.MapTask: mapreduce.local.mapred.mb: 100
2026-06-16 05:05:59.823 INFO mapred.MapTask: mapreduce.local.mapred.mb: 100
2026-06-16 05:05:59.823 INFO mapred.MapTask: bufstart = 0; bufend = 104837600
2026-06-16 05:05:59.823 INFO mapred.MapTask: bufstart = 0; bufend = 104837600
2026-06-16 05:05:59.823 INFO mapred.MapTask: kvstart = 26214396; length = 632200
2026-06-16 05:05:59.823 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-06-16 05:05:59.891 INFO mapred.LocalJobRunner:
2026-06-16 05:05:59.896 INFO mapred.MapTask: Starting flush of map output
2026-06-16 05:05:59.896 INFO mapred.MapTask: Spilling map output
2026-06-16 05:05:59.896 INFO mapred.MapTask: bufstart = 0; bufend = 81; bufend = 104837600
2026-06-16 05:05:59.896 INFO mapred.MapTask: kvstart = 26214396(104837584); kvend = 26214366(104837440); length = 37/653600
2026-06-16 05:05:59.926 INFO mapred.MapTask: Finished spill 0
2026-06-16 05:05:59.930 INFO mapred.Task: Task:attempt_local602077923_0001_n_000000_0 is done. And is in the process of committing
2026-06-16 05:05:59.930 INFO mapreduce.Job: Job job_local602077923_0001 running in uber mode : false
2026-06-16 05:05:59.930 INFO mapreduce.Job: map 0% reduce 0%
2026-06-16 05:05:59.968 INFO mapred.LocalJobRunner: map
2026-06-16 05:05:59.968 INFO mapred.Task: Task:attempt_local602077923_0001_n_000000_0 done.
2026-06-16 05:05:59.983 INFO mapred.Task: Final Counters for attempt_local602077923_0001_n_000000_0: Counters: 24
File System Counters
  FILE: Number of bytes read=3530
  FILE: Number of bytes written=642911
  FILE: Number of read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=0
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=0
  HDFS: Number of write operations=0
  HDFS: Number of bytes read=0
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=0
  HDFS: Number of write operations=0
  HDFS: Number of bytes read=0
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=0
  HDFS: Number of write operations=0
Map-Reduce Framework
  Map input records=1
  Map output records=10
  Map output bytes=89
  Map output materialized bytes=89
  Input split bytes=120
  Combine input records=10
  Combine output records=0
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=9
  Total committed heap usage (bytes)=210763776
File Input Format Counters
  Bytes Read=70
2026-06-16 05:06:00.582 INFO mapred.LocalJobRunner: Finishing task: attempt_local602077923_0001_n_000000_0
2026-06-16 05:06:00.584 INFO mapred.LocalJobRunner: Starting task: attempt_local602077923_0001_n_000001_0
2026-06-16 05:06:00.587 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-16 05:06:00.587 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-16 05:06:00.587 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup, temporary folders under output directory:false, ignore cleanup failures: false
```

```
Reduce shuffle bytes=154
Reduce input records=13
Reduce output records=13
Spilled Records=26
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=22
Total committed heap usage (bytes)=632291328
Shuffle Errors
  BAD_ID=0
  CONNECCIION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
---
File Input Format Counters
  Bytes Read=70
File Output Foomat Counters
  Bytes Written=90
ishwor@OHH:~/wordcount-v1$
```

bytes written: 98

```
a,2
ago,1
another,1
away,1
episode,1
far,2
galaxy,1
in,1
long,1
of,1
star,1
time,1
wars,1
```

PySpark Workshop

Part 6: Spark and CSV Data

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Search:

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☐		Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐		drwxr-xr-x	saksham	supergroup	0 B	Apr 16 10:18	0	0 B	hadoop	
☐		drwxr-xr-x	saksham	supergroup	0 B	Apr 16 10:50	0	0 B	saksham	

Hadoop, 2023.

a	2
ago	1
another	1
away	1
episode	1
far	2
galaxy	1
in	1
long	1
of	1
star	1
time	1
wars	1

```
ishwor@OHH:~/bigdata$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
ishwor@OHH:~/bigdata$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [OHH]
ishwor@OHH:~/bigdata$ ips
```